

LAB IMPLEMENTATION 8

Configuring MetaMask with a DApp

BC

Aim

To configure a cryptocurrency wallet using MetaMask and integrate it with the ChronoVault DApp for wallet connection and blockchain interaction.

Theory

A **cryptocurrency wallet** provides an interface for managing blockchain accounts and approving transactions. MetaMask acts as a browser wallet that can connect a decentralized application (DApp) to an Ethereum-compatible network and exposes the connected account to the application. The DApp can request permissions to view the account and submit transaction requests, while the wallet retains control over transaction approval.

DApp integration generally involves connecting the wallet, selecting an enabled blockchain network, and approving the permissions requested by the application. In this experiment, the ChronoVault DApp uses MetaMask to connect an account and provides workflow options such as TOTP setup, wallet deployment, funding, and transaction execution. The screenshots show the connection request, network permission confirmation, connected-wallet dashboard, account details, and a deployment request on the Sepolia network.

Steps

1. Open the ChronoVault DApp at <https://chronovault-ethonline.vercel.app>.
2. Select MetaMask from the available wallet connection options.
3. Review the MetaMask connection request for the ChronoVault website and inspect the requested account permissions.
4. Continue to the permission review and verify that the requested network is Sepolia.
5. Confirm the wallet connection request in MetaMask.
6. Verify that the ChronoVault dashboard displays the connected wallet address and `Connected` status.
7. Open the account configuration and enter the ChronoVault account details shown in the setup interface.
8. Open the wallet deployment workflow and review the contract deployment request and transaction details presented by MetaMask.

Output

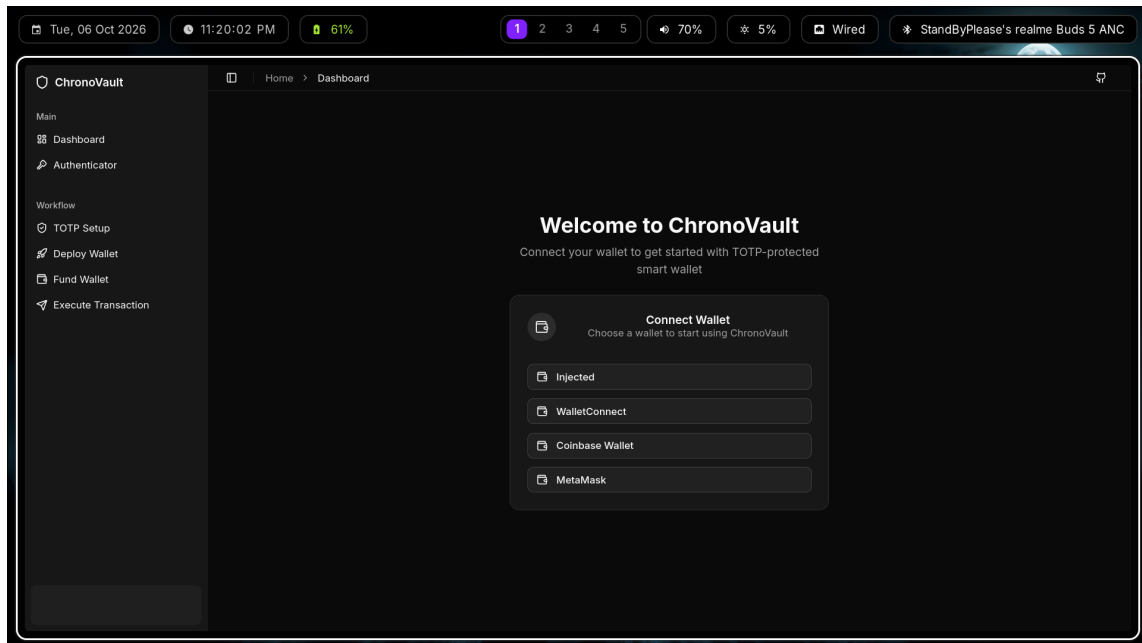


Fig. 1. ChronoVault wallet connection screen

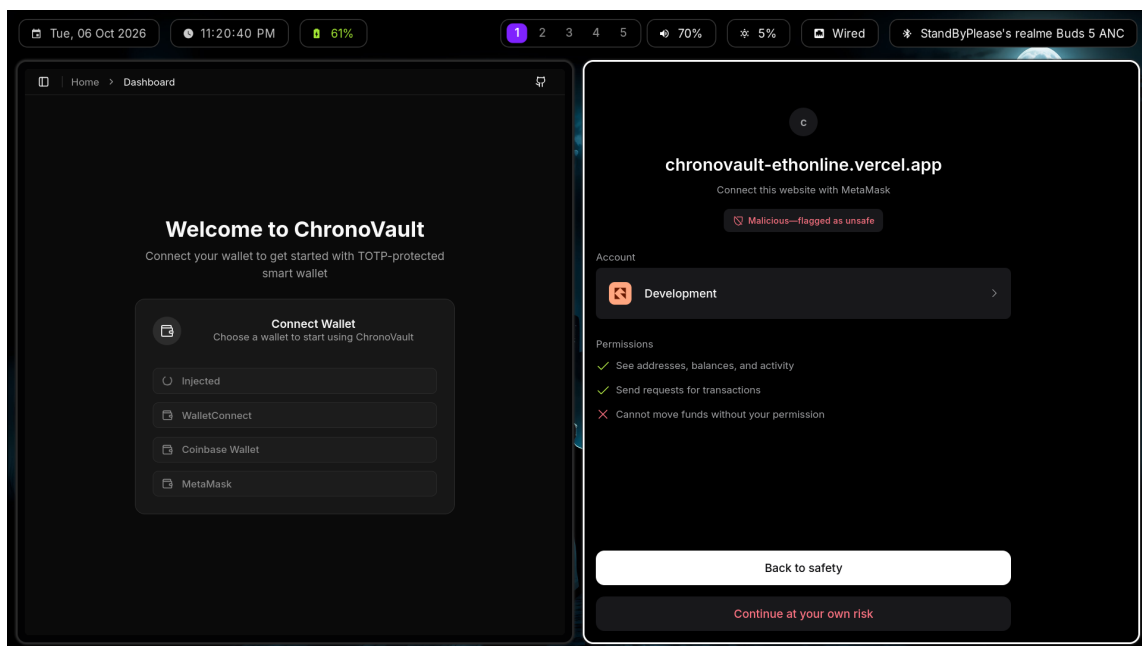


Fig. 2. MetaMask connection request for ChronoVault

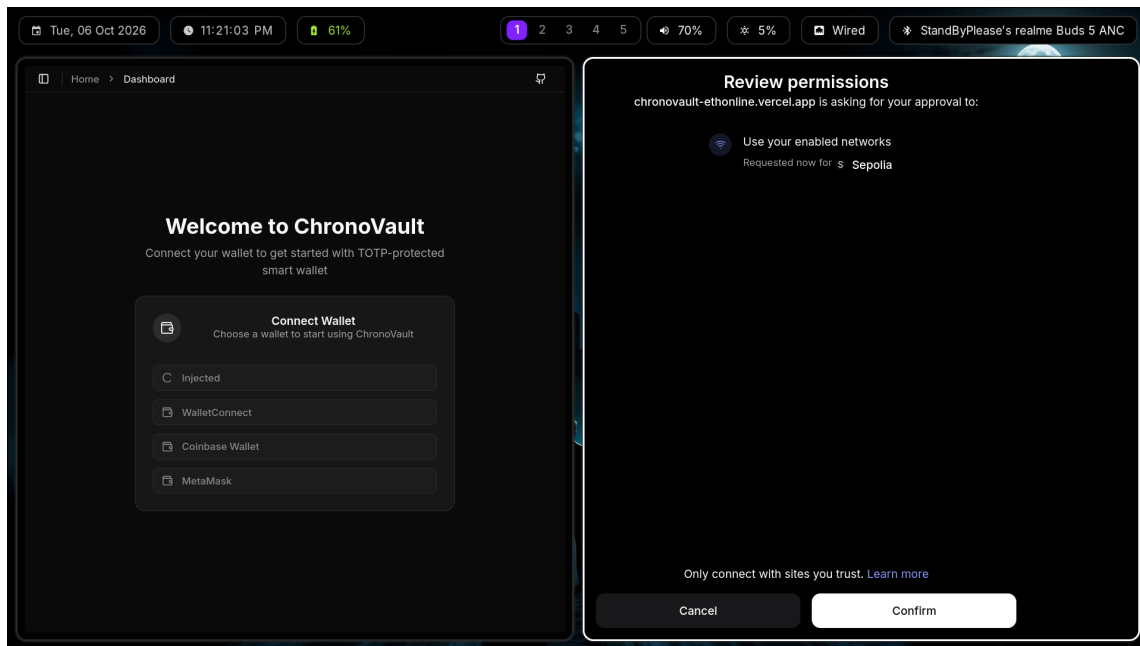


Fig. 3. MetaMask network permission confirmation

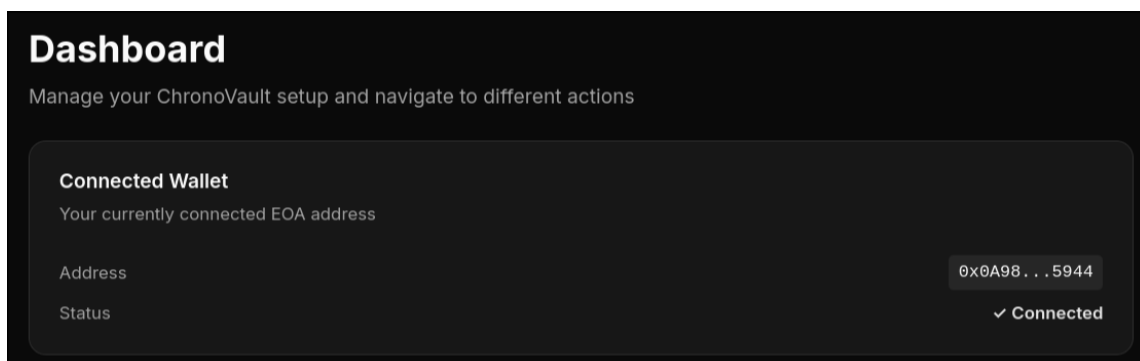
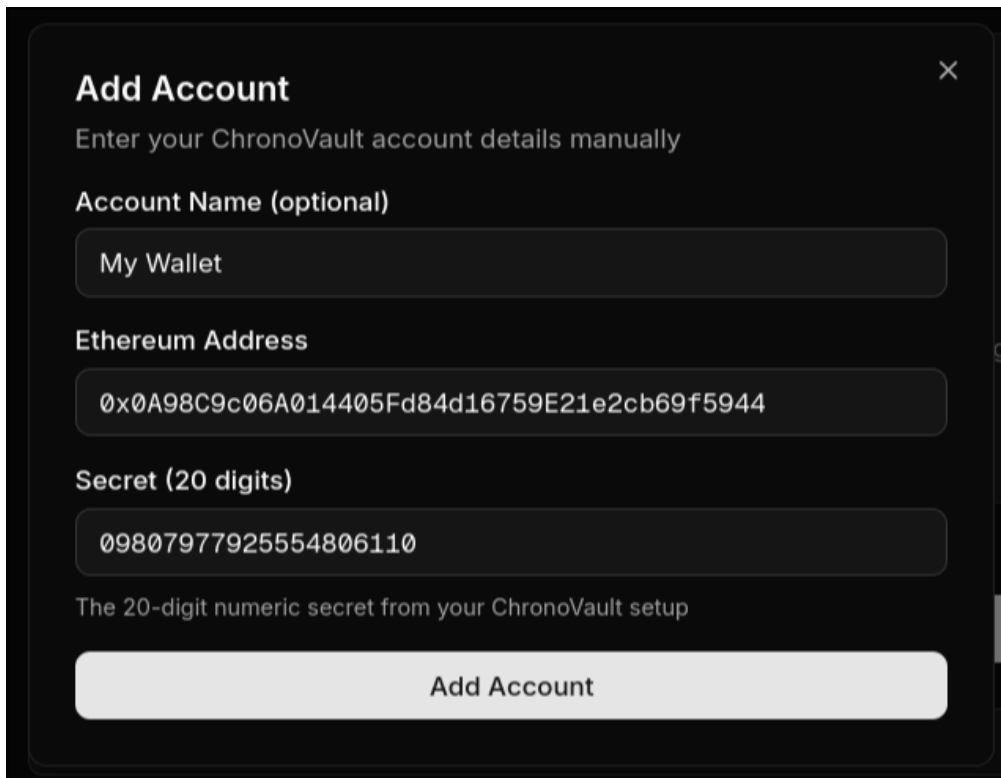
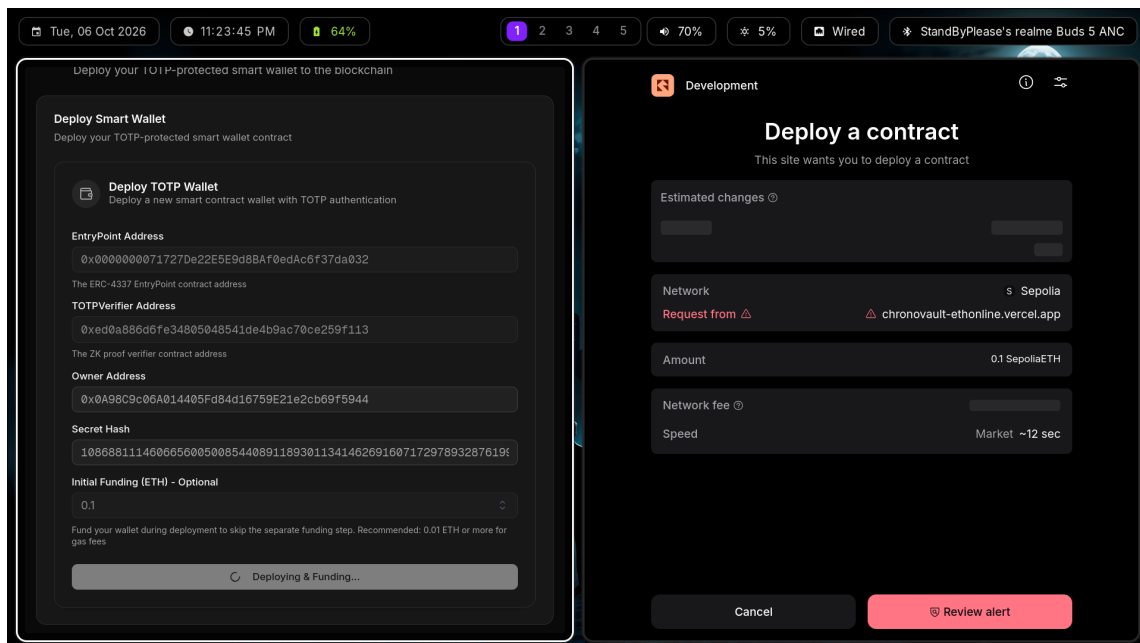


Fig. 4. Dashboard showing the connected wallet



The image shows a dark-themed web form titled "Add Account" with a close button (X) in the top right corner. Below the title is the instruction "Enter your ChronoVault account details manually". The form contains three input fields: "Account Name (optional)" with the text "My Wallet", "Ethereum Address" with the hexadecimal string "0x0A98C9c06A014405Fd84d16759E21e2cb69f5944", and "Secret (20 digits)" with the numeric string "09807977925554806110". A note below the secret field states "The 20-digit numeric secret from your ChronoVault setup". At the bottom is a large, light-colored button labeled "Add Account".

Fig. 5. ChronoVault account details form



The image displays two side-by-side MetaMask interface panels. The left panel, titled "Deploy Smart Wallet", shows fields for "EntryPoint Address", "TOTPVerifier Address", "Owner Address", and "Secret Hash", along with an "Initial Funding (ETH) - Optional" field set to "0.1". A "Deploying & Funding..." button is at the bottom. The right panel, titled "Deploy a contract", shows "Estimated changes", "Network" (Sepolia), "Request from" (chronovault-ethonline.vercel.app), "Amount" (0.1 SepoliaETH), "Network fee", and "Speed" (Market ~12 sec). It includes "Cancel" and "Review alert" buttons at the bottom.

Fig. 6. Smart wallet deployment request in MetaMask

Conclusion

The MetaMask wallet was connected to the ChronoVault DApp and the requested permissions were reviewed for the Sepolia network. The connected-wallet dashboard displayed the wallet connection status, and the account configuration interface was reached successfully. The deployment workflow

also displayed a smart-wallet deployment request through MetaMask. This demonstrated the integration of a browser cryptocurrency wallet with a decentralized application.